SUBJECT - DIGITAL ELECTRONICS AND COMPUTING

Semiconductor Memories

Architecture, classification, and evolution of solid-state digital storage

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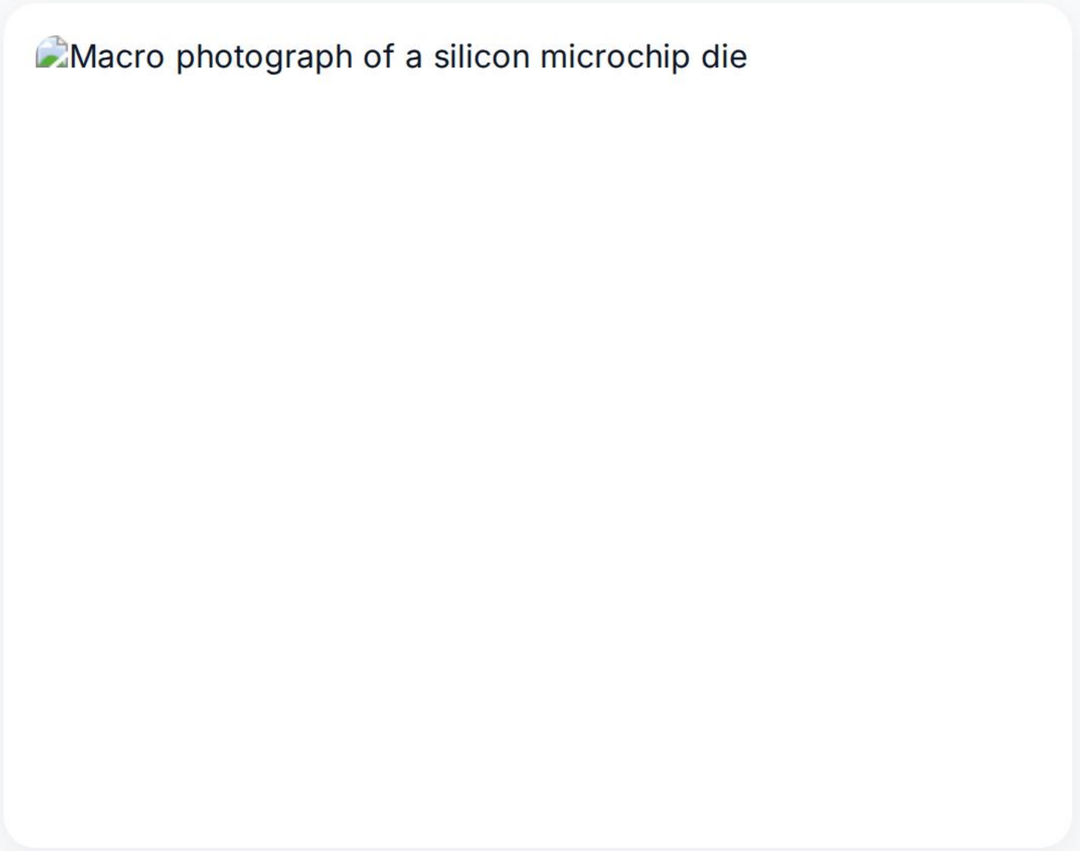
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CORE ARCHITECTURE

Silicon Digital Storage

Semiconductor memories store binary data as electronic states inside integrated silicon circuits.

- **Zero Moving Parts:** Pure solid-state reliability.
- **Nanosecond Access:** Direct electronic read-write speed.
- **High Integration:** Billions of cells per die.



PRIMARY TAXONOMY

Volatile vs. Non-Volatile

Volatile Memory (RAM)

Power Dependency: Data lost without power.

Access Speed: Ultra-fast read and write.

Core Technologies: SRAM and DRAM cells.

Non-Volatile Memory

Data Retention: Persists without electrical power.

Primary Role: Firmware and permanent storage.

Core Technologies: ROM, EEPROM, Flash.

SYSTEM ARCHITECTURE

The Memory Hierarchy

TIER 01 • SUB-1 NS

CPU Registers

Smallest, fastest on-core storage.

TIER 02 • 1–10 NS

SRAM Cache

L1–L3 high-speed processor buffers.

TIER 03 • 50–100 NS

DRAM Main

High-density active system memory.

TIER 04 • MICROSEC

NAND Flash

Massive non-volatile SSD capacity.

VOLATILE • CACHE LAYER

Static RAM (SRAM)

6-Transistor Cell

Cross-coupled flip-flop latch design.

Zero Refresh Needed

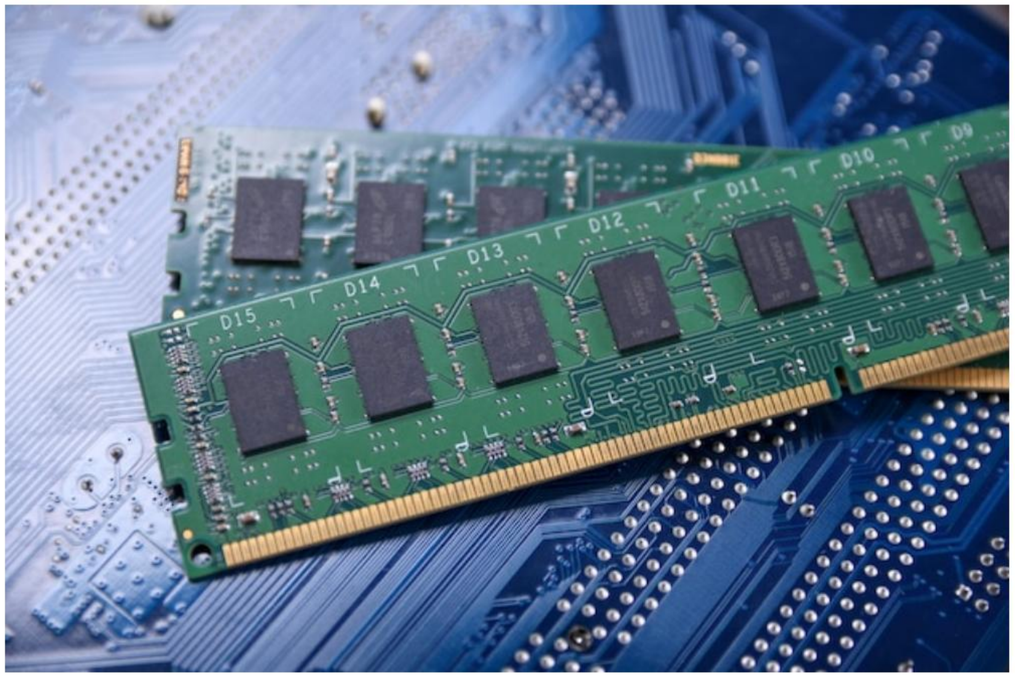
Instantaneous 1–10 nanosecond access.

Processor Cache

Powers CPU registers and L1–L3 caches.

VOLATILE • MAIN MEMORY

Dynamic RAM (DRAM)



Stores each bit as an electrical charge inside a microscopic capacitor paired with one transistor.

- **1T1C Architecture:** Ultra-compact silicon footprint.
- **Periodic Refresh:** Recharged every 64 milliseconds.
- **System Main Memory:** Powers DDR5 PC and server DIMMs.

NON-VOLATILE • FIRMWARE

Read-Only Memory (ROM)

Mask ROM

Hardwired permanently during wafer fabrication.

PROM

One-time user programming via fuse links.

EPROM

Erased using concentrated ultraviolet light.

EEPROM

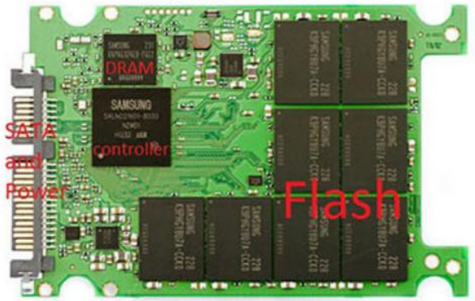
Byte-level electrical erase and rewrite.

NON-VOLATILE • MASS STORAGE

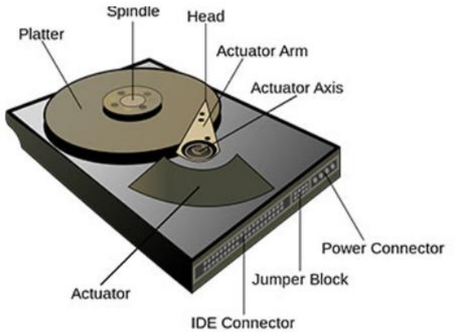
NAND & NOR Flash Memory

Floating-gate and charge-trap transistors retain electrons for years without power.

- **NOR Flash:** Fast random reads for embedded boot code.
- **NAND Flash:** Ultra-high density for SSDs and phones.
- **3D V-NAND:** Vertical cell stacking over 200 layers.



SSD



HDD

NEXT-GENERATION LANDSCAPE

Memory Comparison & Emerging Tech

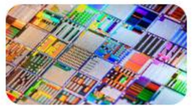
Memory Type	Cell Mechanism	Volatility	Access Speed	Primary Application
SRAM	6-Transistor Latch	Volatile	**1–10 ns**	CPU Cache
DRAM	1T1C Capacitor	Volatile	50–100 ns	Main System RAM
NAND Flash	Charge-Trap Gate	**Non-Volatile**	10–100 µs	SSDs & Mobile
MRAM / ReRAM	Spin / Resistance	**Non-Volatile**	**5–20 ns**	Embedded AI & IoT

Powering Modern Computing

From nanosecond SRAM caches to terabyte 3D NAND arrays, semiconductor memories anchor every digital system.

THANK YOU & QUESTIONS

Image Sources



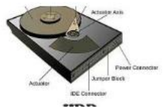
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